

List of final questions:

The list of questions for the final examination in microscopic and gross anatomy including organogenesis for students of general medicine.

Each question covers both microscopic and macroscopic aspects of organ structure, its syntopy, development and the most frequent birth defects

Skeleton and its connections 17- 8

1. Structure and types of bones, innervation and blood supply of bone (draw scheme)
2. Osteogenesis, ossification, remodeling and growth of bone
3. Connection of bones, structure and types of joints
4. The osseous nasal cavity, relations to neighboring structures
5. Bony orbit - walls, relation to neighboring structures, passages
6. Skull, skull of neonate and its development Vertebrae, vertebral column and its development, connections, curvatures and motility
7. Craniovertebral joint
8. Skeleton of thorax and its development, connections and motility of ribs
9. Temporomandibular joint - structure and motility
10. Development and growth of limb, molecular mechanisms, limb defects
11. Shoulder joint - structure and movements Elbow joint - structure and movements
12. Bones and joints of hand including reading of X-ray images
13. Bony pelvis as complex, connections, passages, diameters, planes, sexual differences
14. Hip joint - structure, movements, developmental dysplasia of hip
15. Knee joint - structure, biomechanics, movements
16. Talocrural and subtalar joints - structure, movements
17. Bones and joints of foot including reading of X-ray images, plantar arches and their support

Muscles, fascias, osteofascial compartments 14- 8

18. Origin and development of muscles, molecular mechanisms
19. General features of striated muscle, its auxiliary structures (motor end plate, motor unit, muscle spindle, Golgi tendon organ), motor and proprioceptive innervation
20. Muscles and fascias of the head
21. Muscles and fascias of the neck (draw transverse section of the neck)
22. Muscles and fascias of thorax, diaphragm - structure, passages, function, innervation, diaphragmatic hernias
23. Muscles of abdominal wall, fascias, function
24. Inguinal canal (draw scheme), inguinal hernias
25. Pelvic floor muscles, perineal muscles, ischioanal fossa, pelvic fascias (draw frontal section of pelvis)
26. Muscles and fascias of back
27. Muscles of shoulder girdle, fascias, axillary fossa
28. Muscles and fascias of arm and forearm (draw transverse sections)
29. Muscles and fascias of hand (draw transverse section), tendon sheaths, carpal canal
30. Muscles and fascias of hip Muscles and fascias of thigh, femoral triangle, popliteal fossa (בשאלה 157)
31. Muscles, fascias and compartments of leg and foot (draw transverse sections)

Gastrointestinal tract 26- 8

32. General anatomy (macro and micro) of intestinal tube.
33. Microscopic structure of the teeth and their development
34. Macroscopic structure of the teeth, fixation, gingivodental junction, innervation, and vascular supply,
35. Primary and permanent dentition formula, eruption, types of occlusion
36. Tongue, structure, intra- and extraglosseal muscles, vascular supply, innervation
37. Soft and hard palate, muscles of soft plate (draw scheme), isthmus of fauces
38. Palate development, cleft defects
39. Salivary glands - structure, syntopy, innervation
40. Pharynx - structure, syntopy, blood supply, innervation,
41. Nasal, palatine and lingual tonsils structure (Waldeyer circuit)
42. Oesophagus - structure (macro and micro), syntopy
43. Stomach - shape, position, syntopy, projections

44. Stomach - structure of the wall, divisions, vascular supply, innervation, lymphatic drainage
45. Development of oesophagus, stomach and small and large intestine
46. Small intestine - structure, divisions, vascular supply, innervation, lymphatic drainage
47. Duodenum - divisions, positions, syntopy (draw scheme), blood supply
48. Large intestine, structure, divisions (draw scheme), syntopy, vascular supply, innervation, positions of vermiform appendix
49. Pancreas - structure, Langerhans islets, syntopy,
50. Liver - segments, syntopy (draw scheme of visceral surface)
51. Liver - structure, nutritional and portal vascular bed, intrahepatic bile ducts
52. Gallbladder and extrahepatic bile ducts (draw scheme)
53. Development of pancreas and liver
54. Rectum and anal canal - structure, syntopy (draw frontal and sagittal sections), vascular supply, sphincters, and their innervation.
55. Peritoneum - parietal and visceral, greater and lesser omentum
56. Lesser sac (omental bursa), its recesses
57. Intestinal rotation, development of visceral situs and mesentery

Respiratory system 11- 6

58. Nasal cavity, choanae, vascular and nerve supply
59. Paranasal sinuses and their syntopy, their development
60. Larynx - cartilages, ligaments, joints, muscles (draw frontal section)
61. Larynx - position and syntopy, vascular and nerve supply (draw laryngoscopic view of inlet)
62. Trachea - description , structure, syntopy (draw scheme), tracheotomy
63. Bronchi, bronchial tree - structure, lobar bronchi, segmental bronchi, syntopy
64. Lungs - description, syntopy, bronchopulmonary segments
65. Internal structure of lungs - alveoli and their microscopic structure, surfactant, development, and maturation of lungs
66. Lungs - description, syntopy, borders and projection onto thoracic wall, vascular and nerve supply, lymphatics
67. Pleura - visceral and parietal, structure, borders of pleura, pleural dome and recesses (draw scheme), innervation
68. Mechanics of respiration, pneumothorax

Urinary and reproductive system 19- 9

69. Development of urinary system
70. Kidney - description, position, syntopy (draw scheme), birth defects
71. Structure of kidney - cortex, medulla, nephron, envelopes
72. Vascular supply of kidney, segments
73. Renal calices, pelvis, ureter – syntopy
74. Urinary bladder - structure and position, fixation and syntopy in male and female (draw scheme)
75. Male and female urethra - description, its parts, hypospadias
76. Development of reproductive system
77. Testis and epididymis
78. Scrotum and coverings of testis, descent of testis and its defects
79. Vas (ductus) deferens, spermatic cord, seminal vesicles
80. Prostate - structure, topographic relations, prostatic urethra, ejaculatory ducts
81. Penis - structure (draw cross-section), vascular and nerve supply, mechanism of erection
82. Ovary - structure and position, ovarian cycle, vascular supply, extrauterine gravidity, infertility, IVF
83. Uterine (Fallopian) tube - structure, divisions, position, vascular supply
84. Uterus - shape and divisions, structure of wall, endometrial cycle, vascular supply, lymphatics
85. Uterus - position, fixation, syntopy, birth defects
86. Vagina - structure and syntopy (draw uterus and vagina in sagittal section)
87. External female genital organs, perineum

Heart -8- 7

88. Heart - description, prenatal and postnatal circulation
89. Heart development and its common birth defects
90. Cardiac wall arrangement, cardiac skeleton, chambers (draw section through ventricles)

91. Endocardium, cardiac valves - structure and function, auscultation heart points, cardiac skeleton (draw scheme)
92. Conducting system of the heart- structure and function
93. Coronary arteries, veins and nerves, coronarography
94. Heart location and projection, X-ray (draw scheme of radiogram), auscultation heart points
95. Epicardium and pericardium - structure, syntopy, pericardial reflections around roots of the great vessels, pericardial puncture (pericardiocentesis)

Arteries-9 - 3

96. Vascular development, structure of arteries, veins, lymphatic vessel, collateral circulation
97. Ascending aorta, aortic arch, thoracic aorta (course, syntopy, branches)
98. Common carotid artery, internal carotid artery
99. External carotid artery - course, syntopy, branches
100. Subclavian artery - course, syntopy, branches
101. Arteries of upper extremity - course, syntopy, branches
102. Abdominal aorta, position, topographic relations, parietal and visceral branches
103. External and internal iliac artery Arteries of lower extremity - course, syntopy, branches
104. Anatomical background for vessel punctures, pressure points, palpation

Veins- 7- 2

105. Superior vena cava, brachiocephalic veins, jugular veins
106. Internal jugular vein - course and tributaries
107. Cranial veins, sinus durae matris, cerebral veins
108. Inferior vena cava - course and tributaries, cavocaval anastomoses
109. Azygos and hemiazygos veins, vertebral venous plexuses
110. Portal vein - tributaries, portocaval (portosystemic) anastomosis
111. Superficial and deep veins of upper and lower extremities

Lymphatic system – 9-

112. Thymus - structure, position and syntopy, function
113. Lymph node - structure and functional zones, sentinel lymph node, lymphatic tissue in organs, main lymphatic ducts
114. Spleen - structure, position, syntopy, vascular supply
115. Lymph nodes and collectors of head and neck
116. Lymph nodes and collectors of thoracic wall and lungs
117. Lymph nodes and collectors of stomach, liver and pancreas
118. Lymph nodes and collectors of intestines, lymph nodes and collectors of testis
119. Lymph nodes and collectors of vagina, uterus and ovaries
120. Lymph nodes and collectors of upper and lower limbs

Central nervous system – CNS- 28-

121. Neural tube development and its differentiation, defects of neural tube closure
122. Spinal cord segments, positional changes of cord (cerebromedullary topography), cauda equina
123. Spinal cord - structure of gray and white matter, cross section (draw scheme)
124. Medulla oblongata and pons - gray and white matter, draw cross sections.
125. Floor of rhomboid fossa and cranial nerve nuclei (draw scheme)
126. Mesencephalon (midbrain) - gray and white matter, draw cross section.
127. Reticular formation
128. Cerebellum - structure, subdivision and functional organization, nuclei, and afferent connections
129. Intrinsic and efferent connections of cerebellum and their function
130. Diencephalon - structure, subdivision, and functional organization
131. Thalamus - nuclei, afferent and efferent connections of main nuclei, their function
132. Hypothalamus - subdivisions, connections, and function
133. Anatomical background of hypothalamohypophyseal regulation
134. Basal ganglia, their connections and function, parkinsonism
135. Main functional areas of cerebral cortex

136. White matter of hemispheres - association and commissural fibers, internal capsule (draw scheme of tracts in internal capsule)
137. Ventricular system of brain (draw scheme), circulation of liquor, hydrocephalus
138. Meninges, vascular supply of spinal cord, lumbar puncture
139. Brain vessels and blood-brain barrier, brain dysfunctions related to inadequate blood supply via particular blood vessels brain damage due to vascular occlusion.
140. Lemniscal system (dorsal column tract), proprioceptive and tactile sensation, sensory loss in spinal cord lesions
141. Anterolateral system of sensitive spinal tracts - (spinothalamic, spino- reticular and spino- tectal tracts), pain pathways
142. Corticospinal (pyramidal) and cortico-nuclear tract
143. Motor pathways in spinal cord and motor deficiencies in spinal cord lesions
144. Auditory pathway
145. Visual pathway and visual cortical areas
146. Olfactory and gustatory pathway, olfactory nerve
147. Limbic system - connections and function (cortical areas, hippocampal formation, amygdalar complex)
148. Neurotransmitters in the CNS, main brain chemical systems

Peripheral nervous system - PNS – 18- 8

149. Neural crest cells and their differentiation Ventral and dorsal spinal nerve root, dorsal root ganglion, draw general structure of the spinal nerve and its branches, autonomic fibers of spinal nerve.
150. Segmental innervation, radicular areas, dermatomes, Head's zones (zones of referred visceral pain), sensory receptors, peripheral nerve regeneration
151. Cervical plexus, supraclavicular portion of brachial plexus Infraclavicularis portion of brachial plexus (draw scheme) and upper limb innervation
152. Radial and axillary nerve, paralysis of them
153. Median and ulnar nerve, paralysis of them
154. Skin and motor innervation of head and neck
155. Lumbar plexus and its branches
156. Femoral nerve Sacral plexus and its branches
157. Sciatic nerve, paralysis of common peroneal nerve
158. Overview of muscular and skin innervation of lower limb
159. First and second branch of trigeminal nerve, sensory trigeminal nuclei
160. Third branch of trigeminal nerve
161. Oculomotor, trochlear, abducens nerve
162. Facial nerve Glossopharyngeal and vagus nerve
163. Accessory and hypoglossal nerves
164. Cervical and thoracic sympathetic system
165. Cranial and sacral parasympathetic system
166. Abdominal and pelvic autonomic plexuses and ganglia, enteric system (לחזור אחרי GIT)

Sensory organs, skin, endocrine glands – 15- 3

167. Eye development
168. Eyeball (draw sagittal section)-vascular supply, innervation, chambers, aqueous humor circulation
169. Cornea, sclera and vitreous body, corneal reflex Iris, ciliary body, choroidea, pupillary light reflex
170. Lens - structure and insertion, accommodation
171. Retina - structure, vascular supply (draw schema of eye fundus)
172. Eyelids, conjunctiva, lacrimal apparatus
173. Extraocular muscles
174. External acoustic meatus and tympanic membrane (draw otoscopic view), paracentesis (miringotomy)
175. Tympanic cavity, auditory ossicles, auditory tube
176. Bony and membranous labyrinths (draw cross section of bony cochlea and cochlear duct) vestibulocochlear nerve, nystagmus
177. Skin - epidermis and dermis, skin types, skin receptors, skin appendages (hair, nails, glands)
178. Mamma - description and structure, blood supply and innervation, lymphatics
179. Thyroid and parathyroid glands - structure, function, topography, blood supply
180. Suprarenal gland - structure and developmental origin of cortex and medulla, function, syntopy (draw scheme), blood supply
181. Hypophysis - development, structure of adeno and neurohypophysis, function, syntopy

Regional Anatomy- 30- 24

- 182. Layers of scalp, frontal, and temporal regions
- 183. Superficial regions of face
- 184. Infratemporal fossa and pterygoplatine fossa
- 185. External and internal cranial base - openings for vessels and nerves
- 186. Submandibular triangle, carotid triangle (draw scheme 98 בשאלה 98)
- 187. Lateral neck region, scalenic fissure
- 188. Axilla - boundaries, content
- 189. Brachial region (draw transverse section) 28 בשאלה 28
- 190. Cubital fossa
- 191. Antebrachial region (draw transverse section) 28 בשאלה 28
- 192. Topographic anatomy of the hand (draw transverse section) and fingers
- 193. Gluteal region
- 194. Anterior thigh region, iliopectineal fossa, femoral triangle (draw schema)
- 195. Popliteal fossa (157 בשאלה 157), adductor canal
- 196. Leg region (draw transverse section)
- 197. Retromalleolar regions (med.+ lat.)
- 198. Topography of foot (draw transverse section)
- 199. Mediastinum - division, borders (draw transverse section)
- 200. Topography of chest wall (draw scheme of intercostal space)
- 201. Projections of heart, lungs and pleura onto thoracic wall, puncture of thorax
- 202. Topography of abdominal wall, rectus abdominis sheath (draw transverse section), blood supply, innervation, surgical approaches
- 203. Projections of abdominal organs onto abdominal wall
- 204. Inguinal region, inguinal canal, hernias
- 205. Topography of peritoneal cavity - supra and inframesocolic part (draw transverse section through lesser sac)
- 206. Topography of duodenum and pancreas (draw schema)
- 207. Retroperitoneal space, topography of its organs including vessels and nerves
- 208. Topographic anatomy of male pelvis (draw sagittal section)
- 209. Topographic anatomy of the female pelvis (draw sagittal section), mechanism of continence
- 210. Pelvic floor muscles, ischioanal fossa (draw frontal section of pelvis), perineal region
- 211. Topography of vertebral canal, anatomic backgrounds of spinal tap (lumbar puncture) and epidural anesthesia